

# Advanced Optimization

## Chapter 9 — Derivative-Free Optimization

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Fall 2026

Reading: Nocedal & Wright, *Numerical Optimization*, 2nd ed., Chapter 9

# When derivatives are not available

Typical situations:

- ▶  $f(x)$  is an experimental measurement or a stochastic simulation.
- ▶  $f$  is given only as binary code, or as code in several languages: AD (Chapter 8) cannot be applied.
- ▶ Coding the derivatives is too time consuming.

## The idea

Do not approximate the gradient. Use  $f$  at a set of **sample points** to choose the next iterate.

Methods: model-based (§9.2), coordinate and pattern search (§9.3), conjugate directions (§9.4), Nelder–Mead (§9.5), implicit filtering (§9.6). Not treated: simulated annealing.

## Scope of this chapter

$$\min_{x \in \mathbb{R}^n} f(x) \quad (9.1)$$

- ▶ Unconstrained, and  $f$  has a continuous derivative that we cannot compute.
- ▶ DFO methods are less developed than gradient-based methods: **effective only for small problems**.
- ▶ DFO methods are often used, with mixed success, for nonsmooth or global problems. These are outside the scope of the book.
- ▶ Noise is treated in §9.1 and §9.6.

**Read:** pp. 220–221.

## Noisy function values

The obvious approach: finite-difference gradient + a gradient-based method. It should always be considered, but noise can ruin it. Noise: finite simulation trials, solver tolerances.

$$f(x) = h(x) + \phi(x), \quad h \text{ smooth, } \phi \text{ noise.} \quad (9.2)$$

Central difference (8.7) with interval  $\epsilon$ , and the noise level on the box around  $x$ :

$$\nabla_{\epsilon} f(x) = \left[ \frac{f(x + \epsilon e_i) - f(x - \epsilon e_i)}{2\epsilon} \right]_{i=1, \dots, n}, \quad (9.3)$$

$$\eta(x; \epsilon) = \sup_{\|z-x\|_{\infty} \leq \epsilon} |\phi(z)|. \quad (9.4)$$

## Lemma 9.1: truncation versus noise

### Lemma 9.1

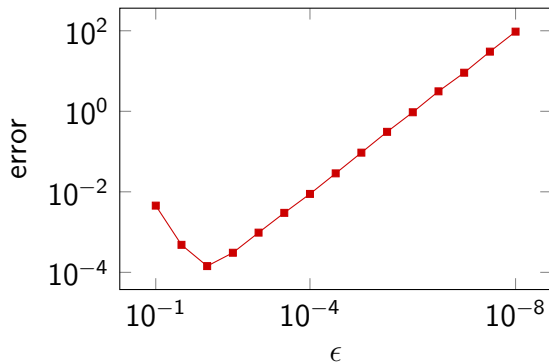
Suppose that  $\nabla^2 h$  is Lipschitz continuous in a neighborhood of the box  $\{z \mid \|z - x\|_\infty \leq \epsilon\}$  with Lipschitz constant  $L_h$ . Then

$$\|\nabla_\epsilon f(x) - \nabla h(x)\|_\infty \leq L_h \epsilon^2 + \frac{\eta(x; \epsilon)}{\epsilon}. \quad (9.5)$$

- ▶ Same argument as for (8.5), with roundoff replaced by noise (Exercise 9.1).
- ▶ If  $\eta$  dominates  $\epsilon$ , the estimate has no accuracy at all. Then  $-\nabla_\epsilon f$  is a descent direction only by luck.
- ▶ Alternative: spread the sample points more widely and build a **model** of  $f$ .

**Read:** pp. 221–223.

## Numerical example: noise moves the best $\epsilon$



$h(x) = e^x$  at  $x = 1$ , uniform noise  $|\phi| \leq \eta = 10^{-6}$ ; central difference, worst of 200 trials.  
(Not in the book.)

- ▶ Best  $\epsilon \approx 10^{-2}$ , error  $\approx 10^{-4}$ .
- ▶ Without noise (Chapter 8): best  $\epsilon \approx 10^{-5}$ , error  $\approx 10^{-10}$ .
- ▶ At  $\epsilon = 10^{-8}$  the “gradient” is off by 100.

## A quadratic model from interpolation

Sample set  $Y = \{y^1, \dots, y^q\}$  containing  $x_k$ , with  $f(x_k) \leq f(y^l)$  for all  $l$ . Model:

$$m_k(x_k + p) = c + g^T p + \frac{1}{2} p^T G p. \quad (9.6)$$

$g$  and  $G$  are **not**  $\nabla f(x_k)$  and  $\nabla^2 f(x_k)$ . They come from interpolation:

$$m_k(y^l) = f(y^l), \quad l = 1, 2, \dots, q. \quad (9.7)$$

Number of coefficients in  $c$ ,  $g$ , symmetric  $G$ :

$$q = \frac{1}{2}(n+1)(n+2). \quad (9.8)$$

The model is usually nonconvex, so the step comes from a trust region:

$$\min_p m_k(x_k + p) \quad \text{subject to} \quad \|p\|_2 \leq \Delta. \quad (9.9)$$

## Algorithm 9.1 (Model-based derivative-free method)

Choose  $Y$  so that (9.7) is nonsingular;  $x_0 \in Y$  with the lowest  $f$ ;  $\Delta_0, \eta \in (0, 1)$ ;  $k \leftarrow 0$ .

**repeat** until a convergence test is satisfied:

Form  $m_k$  satisfying (9.7); compute  $p$  from (9.9);  $x_k^+ = x_k + p$ ;

$$\rho = \frac{f(x_k) - f(x_k^+)}{m_k(x_k) - m_k(x_k^+)}; \quad (9.10)$$

**if**  $\rho \geq \eta$ : replace an element of  $Y$  by  $x_k^+$ ;  $\Delta_{k+1} \geq \Delta_k$ ;  $x_{k+1} \leftarrow x_k^+$ ; next  $k$ ;

**else if**  $Y$  need not be improved:  $\Delta_{k+1} < \Delta_k$ ;  $x_{k+1} \leftarrow x_k$ ; next  $k$ ;

**end if**

*(continued)*

## Algorithm 9.1 (continued)

Geometry improvement: replace at least one point of  $Y$   
to improve the conditioning of (9.7);  $\Delta_{k+1} \leftarrow \Delta_k$ ;  
 $\hat{x} \leftarrow$  element of  $Y$  with the lowest  $f$ ;  $x_k^+ \leftarrow \hat{x}$ ; recompute  $\rho$  by (9.10);  
**if**  $\rho \geq \eta$ :  $x_{k+1} \leftarrow x_k^+$ ; **else**  $x_{k+1} \leftarrow x_k$ ;  
 $k \leftarrow k + 1$ ;  
**end repeat**

When  $\rho < \eta$  there are two possible causes:

- ▶  $Y$  is inadequate: the iterates may be confined to a low-dimensional surface. Detect it by the **conditioning of (9.7)**.
- ▶  $\Delta$  is too large: shrink it, as in Chapter 4.

# The cost of quadratic models

- ▶ Initial  $Y$ : the vertices and the edge midpoints of a simplex in  $\mathbb{R}^n$  (Exercise 9.2).
- ▶  $O(n^2)$  function values just to start: onerous already for  $n = 50$ .
- ▶ Updating  $m_k$  and computing a step:  $O(n^4)$  operations per iteration.

## Linear models

Set  $G = 0$ :  $n + 1$  points and  $O(n^3)$  work per iteration. But linear models cannot capture curvature: **slow convergence**.

Compromise: start with  $n + 1$  points and a linear model; switch to quadratic models once  $q = \frac{1}{2}(n + 1)(n + 2)$  values are available. **Read: pp. 223–226.**

## Interpolation: linear and quadratic models

Linear model, with  $s^l = y^l - x_k$ :

$$m_k(x_k + p) = f(x_k) + g^T p, \quad (s^l)^T g = f(y^l) - f(x_k), \quad l \leq n. \quad (9.11-9.13)$$

Unique iff  $\{s^l\}$  is linearly independent: the simplex  $x_k, y^1, \dots, y^n$  is **nondegenerate**.

Quadratic model in the same linear form:

$$m_k(x_k + p) = f(x_k) + g^T p + \sum_{i < j} G_{ij} p_i p_j + \frac{1}{2} \sum_i G_{ii} p_i^2 \quad (9.14)$$

$$= f(x_k) + \hat{g}^T \hat{p}, \quad (9.15)$$

where  $\hat{g} = (g^T, \{G_{ij}\}_{i < j}, \{G_{ii}/\sqrt{2}\})^T$  (9.16) and

$\hat{p} = (p^T, \{p_i p_j\}_{i < j}, \{p_i^2/\sqrt{2}\})^T$ .

## Polynomial bases and the determinant $\delta(Y)$

The monomial basis (9.14) makes known Hessian structure easy to impose (set  $G_{ij} = 0$ ). For singularity control, other bases are better. With a basis  $\{\phi_i\}_{i=1}^q$  of quadratics,  $m_k(x) = \sum_i \alpha_i \phi_i(x)$ , and  $Y$  determines the  $\alpha_i$  uniquely iff

$$\delta(Y) \stackrel{\text{def}}{=} \det \begin{pmatrix} \phi_1(y^1) & \cdots & \phi_1(y^q) \\ \vdots & & \vdots \\ \phi_q(y^1) & \cdots & \phi_q(y^q) \end{pmatrix} \neq 0. \quad (9.17)$$

**Lagrange functions.** For  $y \in Y$ ,  $L(\cdot, y)$  is quadratic with  $L(y, y) = 1$  and  $L(\hat{y}, y) = 0$  for  $\hat{y} \neq y$ . Replacing  $y_-$  by  $y_+$  gives (after normalization, under conditions)

$$|\delta(Y^+)| \leq |L(y_+, y_-)| |\delta(Y)|. \quad (9.18)$$

# Updating the interpolation set

**Successful step** ( $\rho \geq \eta$ ): add  $x^+$  and remove

$$y_- = \arg \max_{y \in Y} |L(x^+, y)|.$$

**Unsuccessful step** ( $\rho < \eta$ ): is  $Y$  adequate?

- ▶  $Y$  is **adequate** if, for every  $y^i \in Y$  with  $\|x_k - y^i\| \leq \Delta$ ,  $|\delta(Y)|$  cannot be doubled by replacing  $y^i$  with a point in the trust region. Then just shrink  $\Delta$ .
- ▶ Otherwise improve the geometry. For each  $y^i$ , find  $y_r^i = \arg \max_{\|y - x_k\| \leq \Delta} |L(y, y^i)|$ , and remove the  $y^i$  that maximizes  $|L(y_r^i, y^i)|$ .

Implementing these rules well is not simple; see Conn, Scheinberg, Toint [76].

**Read:** pp. 226–228.

## Minimum-change updating

An extension of the quasi-Newton idea (Chapter 6): interpolate only  $\hat{q} = O(n)$  points, and use the remaining freedom to change the Hessian as little as possible.

$$\min_{f, g, G} \|G - G_k\|_F^2 \quad \text{subject to} \quad G \text{ symmetric}, \quad (9.19a)$$

$$m(y^l) = f(y^l), \quad l = 1, \dots, \hat{q}. \quad (9.19b)$$

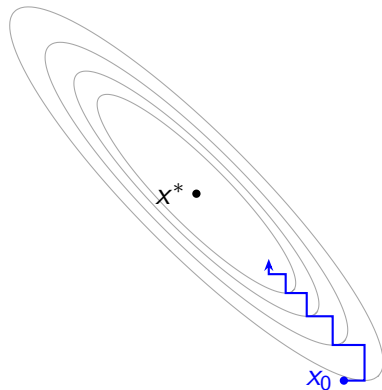
- ▶  $\hat{q} > n + 1$  is needed for  $G_{k+1} \neq G_k$ ; in practice  $\hat{q} = 2n + 1$ .
- ▶ An equality-constrained QP: solve its KKT system. Then a trust-region step (9.9).
- ▶  $O(n^3)$  work per iteration instead of  $O(n^4)$ ; useful steps after only  $O(n)$  evaluations.
- ▶ Implemented in Powell's **NEWUOA**.

**Read:** pp. 228–229.

# Coordinate search

Cycle through  $e_1, e_2, \dots, e_n$ , minimizing (or reducing)  $f$  along each in turn. Also called coordinate descent, or the alternating variables method.

Right:  $f = \frac{1}{2}x^T A x$ ,  $A = \begin{bmatrix} 1 & 0.9 \\ 0.9 & 1 \end{bmatrix}$ ,  
exact line searches. Each cycle reduces the distance to  $x^*$  only by a factor 0.81.  
(Cf. Figure 9.1.)



## Coordinate search: no convergence guarantee

- ▶ It can iterate forever without approaching a stationary point, even with exact line searches. A cyclic search along *any* fixed set of linearly independent directions has this defect.
- ▶ Reason:  $-\nabla f_k$  can become nearly perpendicular to the search direction. Zoutendijk (3.14) then holds because  $\cos \theta_k \rightarrow 0$ , not because  $\|\nabla f_k\| \rightarrow 0$ .
- ▶ When it converges, it is often much slower than steepest descent, and worse as  $n$  grows.
- ▶ Still useful: no gradient, and fast enough if the variables are **loosely coupled**.

Variants: back-and-forth order  $e_1, \dots, e_n, e_{n-1}, \dots, e_1, e_2, \dots$ ; a cycle followed by a search along the line joining its first and last points (Hooke–Jeeves). **Read:** pp. 229–231.

## Pattern search

At  $x_k$ : a direction set  $D_k$  and a step length  $\gamma_k$ . The **frame** (or stencil) is the set of points  $x_k + \gamma_k p_k$ ,  $p_k \in D_k$ .

- ▶ Some frame point gives a significant decrease: move there; possibly enlarge  $\gamma_k$ .
- ▶ None does: stay at  $x_k$  and shrink  $\gamma_k$  (contract the frame).
- ▶  $D_k$  may change between iterations, subject to conditions.

For suitable  $D_k$ , global convergence can be proved: there is a stationary accumulation point.

Noise affects the performance and the theory. Nonsmoothness can cause failure in simple examples, but convergence on nonsmooth problems is often satisfactory in practice.

## Algorithm 9.2 (Pattern search)

Given  $\gamma_{\text{tol}}$ ,  $\theta_{\text{max}} < 1$ , and a sufficient decrease function  $\rho(t)$ , increasing, with  $\rho(t)/t \rightarrow 0$  as  $t \downarrow 0$ ;

Choose  $x_0$ ,  $\gamma_0 > \gamma_{\text{tol}}$ ,  $D_0$ ;

**for**  $k = 1, 2, \dots$

**if**  $\gamma_k \leq \gamma_{\text{tol}}$ : **stop**;

**if**  $f(x_k + \gamma_k p_k) < f(x_k) - \rho(\gamma_k)$  for some  $p_k \in D_k$

$x_{k+1} \leftarrow x_k + \gamma_k p_k$  for some such  $p_k$ ;     $\gamma_{k+1} \leftarrow \phi_k \gamma_k$ ,  $\phi_k \geq 1$ ;

**else**

$x_{k+1} \leftarrow x_k$ ;     $\gamma_{k+1} \leftarrow \theta_k \gamma_k$ ,  $0 < \theta_k \leq \theta_{\text{max}}$ ;

**end for**

Evaluate the frame one point at a time and accept the first point with sufficient decrease. Choice of  $\rho$ :  $\rho \equiv 0$  gives weak theory (Chapter 3); better  $\rho(t) = Mt^{3/2}$ .

## Conditions on the direction set

As in Theorem 3.2, some direction must be close to  $-\nabla f_k$ :

$$\cos \theta = \frac{-\nabla f_k^T p}{\|\nabla f_k\| \|p\|}. \quad (9.20)$$

### Two conditions on $D_k$

$$\kappa(D_k) \stackrel{\text{def}}{=} \min_{v \in \mathbb{R}^n} \max_{p \in D_k} \frac{v^T p}{\|v\| \|p\|} \geq \delta, \quad (9.21)$$

$$\beta_{\min} \leq \|p\| \leq \beta_{\max} \quad \text{for all } p \in D_k. \quad (9.22)$$

Then for any  $\nabla f_k$ , some  $p \in D_k$  satisfies

$$-\nabla f_k^T p \geq \kappa(D_k) \|\nabla f_k\| \|p\| \geq \delta \beta_{\min} \|\nabla f_k\|.$$

# Direction sets that work

Coordinate set and simplex set:

$$\{e_1, \dots, e_n, -e_1, \dots, -e_n\}, \quad (9.23)$$

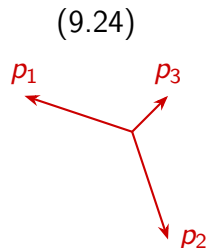
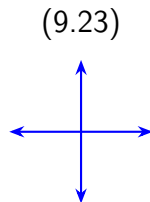
$$p_i = \frac{1}{2n}e - e_i \ (i \leq n), \quad p_{n+1} = \frac{1}{2n}e. \quad (9.24)$$

$e = (1, \dots, 1)^T$ . The simplex set uses only  $n + 1$  directions.

Coordinate descent is the case  $D_k = \{e_i, -e_i\}$ :  
 $\kappa(D_k) = 0$  (Exercise 9.9).

Other directions may be added, from knowledge of  $f$  or from past iterations.

**Read:** pp. 231–234.



$n = 2$  (cf. Figure 9.2)

# The parallel subspace property

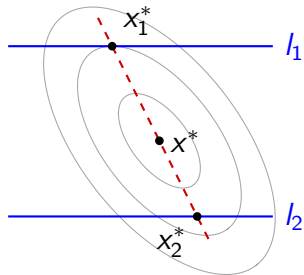
For  $f(x) = \frac{1}{2}x^T Ax - b^T x$  (9.25),  $A \succ 0$ :  
minimize  $f$  along two parallel lines  $x_1 + \alpha p$   
and  $x_2 + \alpha p$ . Then  $x_2^* - x_1^*$  is **conjugate to  $p$** .

Proof:  $\nabla f(x_i^*)^T p = 0$ , so

$$0 = (\nabla f(x_1^*) - \nabla f(x_2^*))^T p = (x_1^* - x_2^*)^T Ap. \quad (9.26)$$

The same holds for parallel affine subspaces  
 $x_i + \text{span}\{p_1, \dots, p_l\}$ .

A line search along  $x_2^* - x_1^*$  reaches  $x^*$  in  $\mathbb{R}^2$ .



$$A = \begin{bmatrix} 2 & 1 \\ 1 & 1.5 \end{bmatrix}, \quad p = e_1 \text{ (cf. Figure 9.3)}$$

## Algorithm 9.3 (DFO method of conjugate directions)

Choose  $x_0$ ;  $p_i = e_i$ ,  $i = 1, \dots, n$ ;  $x_1 \leftarrow$  minimizer of  $f$  along  $x_0 + \alpha p_n$ ;  $k \leftarrow 1$ ;  
**repeat** until a convergence test is satisfied

$z_1 \leftarrow x_k$ ;

**for**  $j = 1, \dots, n$ :  $\alpha_j$  minimizes  $f(z_j + \alpha_j p_j)$ ;  $z_{j+1} \leftarrow z_j + \alpha_j p_j$ ;

$p_j \leftarrow p_{j+1}$ ,  $j = 1, \dots, n - 1$ ;  $p_n \leftarrow z_{n+1} - z_1$ ;

$\alpha_n$  minimizes  $f(z_{n+1} + \alpha_n p_n)$ ;  $x_{k+1} \leftarrow z_{n+1} + \alpha_n p_n$ ;  $k \leftarrow k + 1$ ;

**end repeat**

- ▶ Line searches: quadratic interpolation with three values; exact on (9.25).
- ▶ After iteration  $k$ ,  $p_{n-k}, \dots, p_n$  are conjugate. Termination after  $n - 1$  iterations,  $O(n^2)$  function values, unless a direction becomes zero (Exercise 9.5).

## Keeping the directions independent

On nonquadratic  $f$  the  $p_i$  tend to become linearly dependent. Measure conjugacy by

$$\hat{p}_i = \frac{p_i}{\sqrt{p_i^T A p_i}}, \quad |\det(\hat{p}_1, \dots, \hat{p}_n)|, \quad (9.27, 9.28)$$

maximized iff the  $p_i$  are conjugate (Powell). Keep (9.28) from decreasing:

### Procedure 9.4 (Updating the set of directions)

$m$ : index of the largest decrease  $\psi_m = f(x_{m-1}) - f(x_m)$  in the inner loop;

$f_1 = f(z_1)$ ,  $f_2 = f(z_{n+1})$ ,  $f_3 = f(2z_{n+1} - z_1)$ ;

**if**  $f_3 \geq f_1$  or  $(f_1 - 2f_2 + f_3)(f_1 - f_2 - \psi_m)^2 \geq \frac{1}{2}\psi_m(f_1 - f_3)^2$

    keep  $p_1, \dots, p_n$ ;  $x_{k+1} \leftarrow z_{n+1}$ ;

**else** line search along  $\hat{p} = z_{n+1} - z_1$ ; replace  $p_m$  by  $\hat{p}$ .

**Read:** pp. 234–237.

# The Nelder–Mead simplex method (1965)

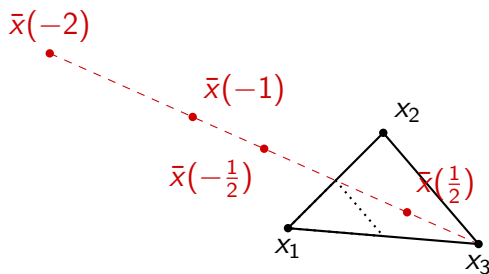
Keep  $n + 1$  points forming a simplex (not the simplex method of Chapter 13), ordered so that

$$f(x_1) \leq f(x_2) \leq \cdots \leq f(x_{n+1}).$$

Centroid of the best  $n$ , and the line through the worst:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad \bar{x}(t) = \bar{x} + t(x_{n+1} - \bar{x}).$$

$t = -1$  reflect,  $-2$  expand,  $-\frac{1}{2}$  outside and  $\frac{1}{2}$  inside contraction.



$n = 2$ ; dotted: shrunken simplex (cf. Figure 9.4)

## Procedure 9.5 (One step of Nelder–Mead)

$f_{-1} = f(\bar{x}(-1));$   
**if**  $f(x_1) \leq f_{-1} < f(x_n)$ : replace  $x_{n+1}$  by  $\bar{x}(-1);$  (reflect)  
**else if**  $f_{-1} < f(x_1)$ :  $f_{-2} = f(\bar{x}(-2));$  (expand)  
    replace  $x_{n+1}$  by  $\bar{x}(-2)$  if  $f_{-2} < f_{-1}$ , else by  $\bar{x}(-1);$   
**else** ( $f_{-1} \geq f(x_n)$ ): (contract)  
    **if**  $f_{-1} < f(x_{n+1})$ : **if**  $f(\bar{x}(-\frac{1}{2})) \leq f_{-1}$ , replace  $x_{n+1}$  by  $\bar{x}(-\frac{1}{2});$  (outside)  
    **else**: **if**  $f(\bar{x}(\frac{1}{2})) < f(x_{n+1})$ , replace  $x_{n+1}$  by  $\bar{x}(\frac{1}{2});$  (inside)  
    if the contraction fails, shrink:  $x_i \leftarrow \frac{1}{2}(x_1 + x_i)$ ,  $i = 2, \dots, n + 1.$

The values  $t = -1, -2, -\frac{1}{2}, \frac{1}{2}$  are standard; other choices are possible.

## Nelder–Mead in practice

- ▶ Unless a shrink step occurs, the average function value over the vertices

$$\frac{1}{n+1} \sum_{i=1}^{n+1} f(x_i) \quad (9.29)$$

decreases at every step (Exercise 9.11). For convex  $f$ , the shrink step does not increase it (Exercise 9.12).

- ▶ Practical performance is often reasonable, but the method can **stagnate at nonoptimal points**. Remedy: restart when stagnation is detected (Kelley).
- ▶ Convergence theory is limited (Kelley; Lagarias et al.).
- ▶ (Not in the book.) McKinnon (1998): a smooth, strictly convex function in  $\mathbb{R}^2$  on which Nelder–Mead converges to a nonstationary point.

**Read:** pp. 238–240.

# Implicit filtering: the idea

For noisy  $f = h + \phi$  as in (9.2):

- ▶ Steepest descent with an Armijo line search (Chapter 3), using the central difference  $\nabla_{\epsilon} f$  (9.3) with an  $\epsilon$  that is **not** particularly small.
- ▶ Works best when the noise decreases near the solution, for example when we control the solver tolerance or the number of simulation trials.
- ▶ Decrease  $\epsilon$  systematically, but more slowly than the noise, so that  $\nabla_{\epsilon} f$  stays accurate (Lemma 9.1).
- ▶ For each  $\epsilon$ : backtracking line search on  $-\nabla_{\epsilon} f$ . If it fails after  $a_{\max}$  backtracks, or if  $\|\nabla_{\epsilon} f\| \leq \epsilon$ , go to the next, smaller  $\epsilon$ .

## Algorithm 9.6 (Implicit filtering)

```
Choose  $\{\epsilon_k\} \downarrow 0$ , Armijo parameters  $c, \rho \in (0, 1)$ ,  $a_{\max}$ ;  $k \leftarrow 1$ ;  $x \leftarrow x_0$ ;  
repeat  
  repeat  
    compute  $f(x)$  and  $\nabla_{\epsilon_k} f(x)$ ;  
    if  $\|\nabla_{\epsilon_k} f(x)\| \leq \epsilon_k$ : leave the inner loop;  
    find the smallest  $m \in \{0, \dots, a_{\max}\}$  with  
       $f(x - \rho^m \nabla_{\epsilon_k} f(x)) \leq f(x) - c \rho^m \|\nabla_{\epsilon_k} f(x)\|^2$ ;  
    if there is none: leave the inner loop; else  $x \leftarrow x - \rho^m \nabla_{\epsilon_k} f(x)$ ;  
  end repeat  
   $x_k \leftarrow x$ ;  $k \leftarrow k + 1$ ;  
until a termination test is satisfied           (inner loop = Algorithm 3.1 + a test)
```

## Theorem 9.2: convergence of implicit filtering

### Theorem 9.2

Suppose that  $\nabla^2 h$  is Lipschitz continuous, that Algorithm 9.6 generates an infinite sequence  $\{x_k\}$ , and that

$$\lim_{k \rightarrow \infty} \left( \epsilon_k^2 + \frac{\eta(x_k; \epsilon_k)}{\epsilon_k} \right) = 0.$$

Suppose also that all but finitely many inner loops terminate with  $\|\nabla_{\epsilon_k} f(x_k)\| \leq \epsilon_k$ . Then all limit points of  $\{x_k\}$  are stationary.

**Proof.**  $\epsilon_k \downarrow 0$  gives  $\nabla_{\epsilon_k} f(x_k) \rightarrow 0$ . The right-hand side of (9.5) tends to 0, so  $\nabla h(x_k) \rightarrow 0$ . □

The noise must vanish faster than  $\epsilon_k$ . Refinement: use  $\nabla_{\epsilon_k} f$  in quasi-Newton updates. **Read:** pp. 240–242.

## Numerical example: Rosenbrock's function (2.22)

Starting point  $(-1.2, 1)$ ; SciPy 1.17, default settings with tight tolerances. (Not in the book.)

| Method                                            | Section | $f$ evaluations      | $\ x - x^*\ $       |
|---------------------------------------------------|---------|----------------------|---------------------|
| Nelder–Mead                                       | §9.5    | 219                  | $2 \times 10^{-9}$  |
| Powell (conjugate directions)                     | §9.4    | 792                  | $5 \times 10^{-14}$ |
| Pattern search (9.23), $\rho(t) = 10^{-4}t^{3/2}$ | §9.3    | 19 182               | $9 \times 10^{-6}$  |
| COBYLA (linear models)                            | §9.2    | 18 294               | $3 \times 10^{-7}$  |
| BFGS + forward differences                        | §9.1    | 120                  | $1 \times 10^{-5}$  |
| BFGS + exact gradient                             | Ch. 6   | 41 (+41 $\nabla f$ ) | $1 \times 10^{-12}$ |

Linear models and fixed directions are slow on a curved valley. Finite differences are cheap but stop at about  $\sqrt{u}$  accuracy.

## Chapter 9 in one table

| Method                     | Uses                             | Theory                             |
|----------------------------|----------------------------------|------------------------------------|
| FD + gradient              | tight cluster                    | fails if $\eta \gg \epsilon$ (9.5) |
| Model-based (9.1)          | interpolation + TR               | geometry of $Y$                    |
| Minimum change (9.19)      | $2n + 1$ points, $O(n^3)$        | NEWUOA                             |
| Coordinate search          | $\pm e_i$ , cyclic               | may not converge                   |
| Pattern search (9.2)       | frame, $\kappa(D_k) \geq \delta$ | stationary limit point             |
| Conjugate directions (9.3) | parallel subspaces               | finite on quadratics               |
| Nelder–Mead (9.5)          | simplex reflection               | limited; can stagnate              |
| Implicit filtering (9.6)   | FD with shrinking $\epsilon_k$   | Theorem 9.2                        |

# Three results to carry forward

1. **Noise sets the difference interval.**  $\text{Error} \leq L_h \epsilon^2 + \eta/\epsilon$  (Lemma 9.1): with noise, small  $\epsilon$  is harmful.
2. **Model-based DFO is trust-region optimization with an interpolated model.** The new ingredient is keeping the interpolation set well poised:  $\delta(Y)$ , Lagrange functions.
3. **Directions must cover all of  $\mathbb{R}^n$ .** Pattern search converges because  $\kappa(D_k) \geq \delta > 0$ ; coordinate descent has  $\kappa = 0$ .

Use DFO only for small  $n$ , and only when derivatives (Chapter 8) are truly unavailable.

## Homework (problems 5–6 are not in the book)

1. Exercise 9.1: prove Lemma 9.1.
2. Exercise 9.2: count the interpolation conditions; vertices and edge midpoints of a simplex.
3. Exercise 9.7: find the quadratic that interpolates the six given data points.
4. Exercise 9.9: show that  $\kappa(\{e_i, -e_i\}) = 0$ .
5. Implement Algorithm 9.2 with the sets (9.23) and (9.24). Run it on Rosenbrock (2.22) from  $(-1.2, 1)$  and compare the number of evaluations.
6. Add uniform noise of size  $10^{-4}$  to Rosenbrock. Compare Nelder–Mead with BFGS using forward differences,  $\epsilon = \sqrt{u}$  and  $\epsilon = 10^{-2}$ .